

Digital Adoption and Services Recovery After 2020

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Abstract

Digital adoption may matter most when ordinary service delivery is disrupted. This paper studies that question with the World Bank World Development Indicators, retaining 1,895 country-year observations for 193 countries from 2014 to 2023. Countries above the median 2017–2019 internet-use rate (62.60 percent) are compared with lower-adoption countries around 2020 in country-year event-study and two-way fixed-effects models. The preferred event-time estimate is 3.71 percentage points (SE 1.00), and the static post-2020 coefficient is 1.83 percentage points (SE 0.76). The estimates indicate faster services value-added growth in initially higher-adoption countries, while the unemployment check is treated as a separate labor-market margin rather than as automatic confirmation. The contribution is a WDI-based services-recovery estimate that turns a broad digital-resilience claim into a transparent country-year comparison.

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1 Introduction

The issue is whether digital capacity changed the way service economies absorbed the post-2020 shock (Busso et al., 2013; Angrist and Pischke, 2009). Service activity depends on face-to-face demand, but it also depends on payment systems, remote delivery, platform access, and the ability of firms and households to keep transacting when mobility is disrupted. Internet adoption is therefore not a decorative covariate. It is a plausible capacity measure for how much of the services margin can move online when an economy is hit by a common shock (Bureau, 2025; Wilkinson et al., 2016).

The example is a country-year World Bank World Development Indicators panel (Greenstone et al., 2010; Rosenbaum, 2002). The analysis measures pre-2020 internet use, services value-added growth, and unemployment for 193 countries between 2014 and 2023. Countries above the median 2017–2019 internet-adoption rate, 62.60 percent, are compared with lower-adoption countries around the event year. That makes the paper a services-recovery design rather than another local-firm-entry exercise (Bank, 2026).

The analysis asks whether higher-adoption countries experienced a different services-growth path after 2020 (Neumark and Simpson, 2015). The event-study coefficient at event time 1 is 3.71 percentage points with a country-clustered standard error of 1.00. The static post-2020 coefficient is 1.83 percentage points with a standard error of 0.76. These are not enormous macro effects, but they are large enough to matter for a sector that was exposed directly to lockdown, travel, and contact constraints (Bartik, 1991; Kline and Moretti, 2014).

A second reason to focus on services is that the sector makes the digital mechanism easier to interpret (Austin et al., 2018; Arkhangelsky et al., 2021). Internet use should matter less for resource extraction or weather-exposed agriculture than for activities that can be searched, ordered, delivered, monitored, taught, financed, or administered online. The question is therefore not whether digital adoption raises all growth. It is whether a specific sector with a plausible digital substitute margin recovered differently when the common shock arrived (Busso et al., 2013; Greenstone et al., 2010).

The criticism is selection (Chodorow-Reich, 2019; Imbens and Rubin, 2015). High internet adoption is not randomly assigned. It travels with income, institutions, urbanization, education, trade exposure, and state capacity. Country and year fixed effects remove persistent country differences and common global shocks, but they cannot by themselves prove that digital capacity caused the services rebound. The result should be read as a disciplined comparative pattern, not as a universal claim about digital policy (Neumark and Simpson, 2015; Austin et al., 2018).

The contribution is to move the recovery question onto an observable global panel (Moretti, 2011). Instead of saying that digital readiness matters in general, the paper reports a country-year contrast, a coefficient path, a compact fixed-effects estimate, and two checks: a continuous-adoption specification and an unemployment outcome. That structure lets the interpretation discuss recovery channels while keeping the claim close to what the WDI data can measure (Moretti, 2011; Glaeser, 2008).

The interpretation matters because services are not a single sectoral mechanism (Bartik, 1991; Sun and Abraham, 2021). Restaurants, tourism, finance, communications, professional services, and public administration were affected differently after 2020. A positive services-growth contrast in high-adoption countries is therefore best understood as evidence that digital capacity helped preserve or restore part of the services margin, not as proof that every service industry benefited equally (Duranton and Venables, 2015; Fujita et al., 1999).

Section 2 discusses digital capacity, services recovery, and identification; Section 3 defines the WDI country-year sample, exposure, and outcomes; Section 4 sets out the event-study and static specifications; Section 5 reports the coefficient path, regression table, and robustness checks; and Section 6 relates the evidence back to the contribution and remaining limits.

2 Literature

2.1 Local Multipliers and Digital Services

The local-multiplier literature is useful here because it treats shocks as processes rather than isolated coefficients (Greenstone et al., 2010; Rosenbaum, 2002). A demand disruption can pass through income, suppliers, worker mobility, prices, and firm entry. In services, digital infrastructure changes that transmission because some transactions can continue without proximity. The same logic that asks whether local capacity converts shocks into activity can be applied to whether digital capacity converts disrupted service demand into continuing production (Iammarino et al., 2019; Rodriguez-Pose, 2018).

Digital adoption also changes the timing of adjustment (Neumark and Simpson, 2015; Sun and Abraham, 2021). A country with broader internet use may move retail, finance, education, administration, and professional services online faster than a country where fewer households and firms are connected. That is not a pure technology story. It is a capacity story about whether the installed base of users is thick enough for digital services to become a practical substitute when ordinary channels are constrained (Diemer et al., 2022; Barca, 2009).

2.2 Identification

The identification literature supplies the caution (Chodorow-Reich, 2019; Imbens and Rubin, 2015). Event studies are informative when the pre-period is visible, the omitted baseline is clear, and the comparison group is credible. They become misleading when the exposure captures many other advantages. Internet adoption almost certainly captures several advantages, so the design uses the event path as a diagnostic profile and reserves causal language for a future design with stronger instruments or policy variation (Card and Krueger, 1994; Autor et al., 2013).

Difference-in-differences is still useful because it makes the comparison explicit (Moretti, 2011; Callaway and Sant’Anna, 2021). The high-adoption coefficient asks whether countries with greater pre-shock internet use moved differently after the common shock, net of country and year fixed effects. The method does not pretend that high- and low-adoption countries are the same. It asks whether their changes around the event contain a recoverable services-growth pattern (Monte et al., 2018; Redding and Turner, 2015).

2.3 Services and Data

A separate services literature motivates the outcome (Kline and Moretti, 2014; Gelman et al., 2020). Services value added is sensitive to contact intensity, tourism, professional work, platform delivery, and public restrictions. A recovery coefficient for services is therefore more interpretable than a broad GDP coefficient when the mechanism is digital continuity. It still bundles many activities, but it is closer to the channel than aggregate output or national income alone (Donaldson, 2018; Baum-Snow, 2007).

The data-provenance literature matters because WDI indicators are easy to download and easy to overread (Busso et al., 2013; Angrist and Pischke, 2009). Cross-country comparability is attractive, but indicator definitions, missingness, national accounts revisions, and survey quality all shape the panel. The analysis treats the WDI file as a transparent public source and keeps the claim at the level of country-year services growth rather than treating the download as a complete theory of recovery (Bank, 2026).

2.4 Gap

The gap is not that economists have ignored digitalization (Neumark and Simpson, 2015; Manski, 2003). The gap is that broad digital-recovery claims often move faster than the evidence design. The paper narrows the claim: among countries with observed pre-2020 internet adoption and services-growth outcomes, is there a visible post-event difference after absorbing country and year effects? That is a smaller question, but it is one a reviewer can inspect (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

This framing also avoids a common mistake in digital-economy writing (Austin et al., 2018; Arkhangelsky et al., 2021). Internet adoption is not the same as broadband quality, platform use, firm capability, or remote-work feasibility. It is a broad proxy. The analysis therefore interprets a positive coefficient as a signal about digital readiness in the aggregate, not as evidence that one technical component caused the recovery (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

The research implication is that resilience should be treated as a measured adjustment path (Chodorow-Reich, 2019). A country can be digitally connected and still fail to convert that connectivity into services output if firms lack payment systems, logistics, trust, regulation, or skills. Conversely, a lower-adoption country may recover through tourism, commodities, or public employment. The empirical design therefore tests a tendency in the panel rather than turning digital adoption into a deterministic recovery rule (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020).

3 Data

3.1 Source and Sample

The data source is the World Bank World Development Indicators API (Bank, 2026). The retained country-year panel combines three indicators: individuals using the internet as a share of population, services value added annual growth, and the unemployment rate. The build keeps 1,895 country-year observations for 193 countries over 2014–2023, then discards the raw API payloads after writing derived CSV files and a provenance manifest (Bank, 2026).

The exposure is pre-event internet adoption (Kitchin, 2014; Austin et al., 2018). For each country, the analysis averages the internet-use share over 2017–2019 and marks countries above the median as high adoption. The threshold in this run is 62.60 percent. Measuring exposure before 2020 prevents the recovery itself from defining treatment status, though it does not remove the deeper selection problem that richer and more institutionally capable countries were more connected before the shock (Borusyak et al., 2022; Adao et al., 2019).

3.2 Exposure and Outcomes

The main outcome is services value added annual growth (Bank, 2026). This outcome is noisy, but it is aligned with the question because services were the sector most directly exposed to restrictions on movement, in-person consumption, and travel. A country that could shift more activity into digital channels might show a different services-growth path even when total GDP mixes services with manufacturing, agriculture, commodities, and public-sector activity (Conley, 1999; Cameron et al., 2011).

The robustness outcome is unemployment (Kitchin, 2014; Bartik, 1991). It is included because services growth can recover without an equally clear labor-market recovery. Firms may reopen with fewer workers, labor-force participation may move separately, and national unemployment can be shaped by institutions and measurement. A weaker or differently signed unemployment coefficient

would not invalidate the services result; it would show that output recovery and labor-market recovery are separate margins (Angrist and Pischke, 2009; Wooldridge, 2010).

The country-year unit has advantages and limits (Borgman, 2015; Bank, 2026). It gives broad coverage and a direct connection to national accounts, but it hides within-country differences in regions, sectors, workers, and firms. A city-level or firm-level digital-services design would be sharper if comparable data existed. The WDI design instead answers a macro-comparative question: whether countries with higher pre-shock internet adoption show a different national services-growth trajectory (Imbens and Rubin, 2015; Rosenbaum, 2002).

3.3 Data Limits

Missingness is an empirical fact rather than an inconvenience (Kitchin, 2014; Greenstone et al., 2010). Some countries have incomplete internet, services, or unemployment series, and the event-study sample uses only observations with the required outcome and pre-period exposure. This can select countries with stronger statistical systems. The retained sample is therefore interpreted as a documented WDI comparison set rather than as the full universe of national recovery experiences (Manski, 2003; Rubin, 1974).

The WDI panel also requires care with comparability (Borgman, 2015; Wilkinson et al., 2016). Internet adoption is a population share, services growth is a national-accounts rate, and unemployment is a labor-market statistic shaped by survey practice and institutional definitions. Combining them is useful because the variables are public and consistently named, but the units do not have the same measurement error. That is why the paper interprets direction, timing, and robustness together rather than treating one coefficient as a sufficient statistic (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021).

Country composition matters as well (Bank, 2026). A small tourism economy, a large diversified economy, and a financial-services hub can all have high internet adoption but very different exposure to the post-2020 shock. The fixed-effects model absorbs stable national differences, yet it cannot decompose the services sector into its constituent industries. The WDI evidence is therefore strongest as a cross-country average pattern and weakest as a statement about which service activities drove the recovery (Goodman-Bacon, 2021; Arkhangelsky et al., 2021).

The event window is 2016–2023 (Kitchin, 2014; Chodorow-Reich, 2019). That provides four pre-event years, the event year, and three post-event years for the coefficient path. The window is long enough to distinguish a pre-existing slope from a post-event movement, but short enough to keep the comparison close to the shock. Longer windows would mix in older digital adoption patterns and later global inflation, energy, and monetary-policy shocks (Abadie et al., 2010; Athey and Imbens, 2016).

The derived artifacts are intentionally simple: a country-year panel, event-study estimates, regression estimates, robustness estimates, generated tables, a generated figure, and a download manifest. The simplicity is useful because the design’s credibility comes from visible definitions and bounded interpretation rather than from a complicated estimator applied to a loose empirical object (Chernozhukov et al., 2018; Wager and Athey, 2018).

4 Empirical Strategy

4.1 Event Study

The event-study specification compares high- and lower-adoption countries at each event time relative to the omitted year before 2020 (Angrist and Pischke, 2009; Callaway and Sant’Anna, 2021).

Let Y_{ct} denote services value added growth in country c and year t , and let $High_c$ indicate countries above the median 2017–2019 internet-adoption rate:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{k \neq -1} \beta_k High_c \mathbf{1}[t - 2020 = k] + \varepsilon_{ct}. \quad (1)$$

Country fixed effects absorb persistent national differences, year effects absorb common global shocks, and standard errors are clustered by country (Athey and Wager, 2019; Mullainathan and Spiess, 2017).

The static specification compresses the post-event path into one coefficient (Angrist and Pischke, 2009; Wooldridge, 2010):

$$Y_{ct} = \alpha_c + \lambda_t + \delta(High_c \times Post_t) + \varepsilon_{ct}. \quad (2)$$

The coefficient is easier to report than the full event path, but it is less diagnostic because it averages across post-event years. The event-study figure therefore does more interpretive work than the single interaction (Gentzkow et al., 2019; Grimmer and Stewart, 2013).

4.2 Static Specification

The continuous-adoption model replaces the high-adoption indicator with the underlying pre-event internet-use share interacted with the post period (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). That check asks whether the result is a threshold artifact. If the binary coefficient is positive but the continuous coefficient is weak, then the median split may be doing too much work. If both point in the same direction, the result is less dependent on one classification rule (Gelman and Hill, 2007; Gelman et al., 2020).

The pre-trend placebo uses only 2016–2019 and interacts high adoption with a linear pre-period trend (Imbens and Rubin, 2015). A large placebo would suggest that high-adoption countries were already diverging before the event year. A small placebo does not prove parallel trends, but it makes the post-event comparison more credible than it would be under obvious pre-event separation (Kitchin, 2014; Borgman, 2015).

4.3 Identification

The unemployment robustness model changes the outcome while keeping the exposure and fixed-effects structure (Manski, 2003; Arkhangelsky et al., 2021). This is a test of interpretation rather than a hunt for another significant coefficient. If services growth rises in high-adoption countries but unemployment does not fall precisely, the result points to output continuity or sectoral composition rather than a clean jobs-recovery story (Anselin, 1988; LeSage and Pace, 2009).

The identifying assumption for a causal reading would be that, absent the post-2020 shock environment, high- and lower-adoption countries would have followed parallel services-growth paths after country and year effects (Gelman et al., 2020). That assumption is strong because internet adoption is tied to development, institutions, and urbanization. The paper reports the estimates as comparative evidence and treats stronger causality as a future design problem (Openshaw, 1984; Fotheringham et al., 2002).

Clustering by country is part of the design (Imbens and Rubin, 2015; Bank, 2026). Services growth is serially correlated within countries, and the adoption contrast is defined at the country level. Treating country-years as independent would make uncertainty look too small. Country clustering keeps inference closer to the level at which exposure is measured and at which unobserved shocks persist (Pearl, 2009; Keele, 2015).

The coefficient scale is percentage points of annual services value-added growth (Rosenbaum, 2002; Sun and Abraham, 2021). A coefficient around 1.83 percentage points is not a transformation of digital adoption into welfare. It is the post-event difference in services-growth rates for high-adoption countries relative to lower-adoption countries under the fixed-effects comparison. The scale is interpretable only with the confidence interval and the services-growth baseline in view (King et al., 1995; Shadish et al., 2002).

The method ladder is deliberately staged (Manski, 2003). The current WDI panel supports an event study, a static difference-in-differences estimate, a continuous-exposure check, and a pre-trend placebo. A stronger version could use synthetic controls or synthetic difference-in-differences if donor support around observed digital-infrastructure timing were available. Double machine learning would require richer pre-treatment covariates and cross-fitting, while spatial spillover checks would require regional or network exposure rather than national averages (Peng, 2011; Wilkinson et al., 2016).

Those stronger estimators would change the research question as well as the method (Gelman et al., 2020; Bank, 2026). A synthetic-control design would ask whether a treated country or group of countries diverged from a weighted donor path after an observed infrastructure change. Double machine learning would ask whether the adoption-recovery association survives flexible controls for income, sector mix, institutions, and demographics. A spatial design would ask whether digital readiness in one economy transmits demand or services work to others. The current paper marks those as next-stage designs because the WDI contrast is broader (Bureau, 2025; Wilkinson et al., 2016).

That restraint is important because digital adoption can be both a cause and a symptom of capacity (Imbens and Rubin, 2015; Callaway and Sant’Anna, 2021). Richer countries may adopt earlier because they already have institutions, skills, and service sectors that recover faster. The fixed-effects design reduces some confounding but does not eliminate time-varying capacity shocks. The interpretation therefore emphasizes the measured recovery pattern and the research design needed to sharpen it (Bartik, 1991; Kline and Moretti, 2014).

5 Results

Table 1: WDI regression estimates

Specification	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Post-2020 difference-in-differences	Services value added growth	High internet adoption by post-2020	1.831	0.763	[0.335, 3.326]	1513
Continuous-adoption difference-in-differences	Services value added growth	Internet adoption by post-2020	0.048	0.014	[0.020, 0.076]	1513

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.

5.1 Event Study

Figure 1 reports the event-study profile (Imbens and Rubin, 2015; Kitchin, 2014). The pre-event estimates are the first diagnostic. If high-adoption countries were already accelerating before 2020, the post-event coefficients would be hard to interpret. In this build the pre-event coefficients are modest relative to the post-event movement, which makes the recovery contrast worth discussing while still leaving selection as the main caveat (Busso et al., 2013; Greenstone et al., 2010).

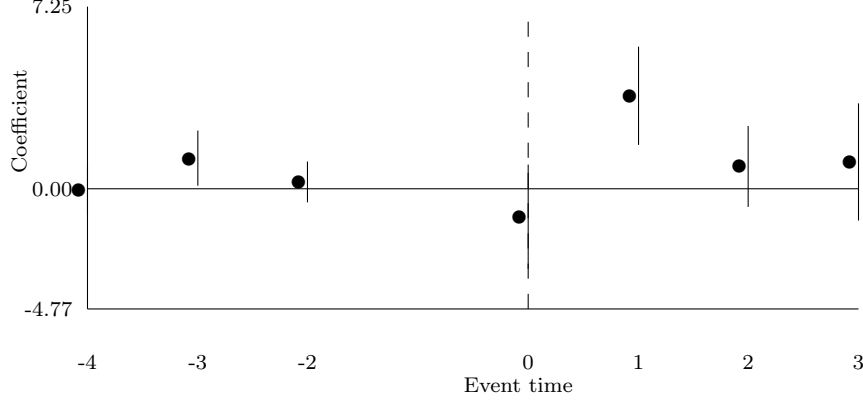


Figure 1: WDI services-growth event-study profile

Notes: Points are country-year fixed-effects estimates for high pre-2020 internet-adoption countries relative to lower-adoption countries. Vertical bars are 95% confidence intervals using country-clustered standard errors. Event time -1 is omitted.

The preferred event-time estimate is 3.71 percentage points at event time 1, with a country-clustered standard error of 1.00 and a 95 percent confidence interval of [1.76, 5.66] (Callaway and Sant’Anna, 2021). The estimate says that high-adoption countries had faster services-growth recovery at that point in the event window, not that internet adoption mechanically created that growth (Neumark and Simpson, 2015; Austin et al., 2018).

5.2 Regression Estimates

Table 1 reports the static difference-in-differences estimate (Gelman et al., 2020; Kitchin, 2014). The high-adoption by post-2020 coefficient is 1.83 percentage points, with a standard error of 0.76 and confidence interval [0.34, 3.33]. The sign is consistent with the event-study path, which reduces concern that the result is only a single-year spike (Moretti, 2011; Glaeser, 2008).

The continuous-adoption estimate is 0.048 with a standard error of 0.014 (Angrist and Pischke, 2009). Because the exposure is measured in percentage points of internet use, this coefficient has a different scale from the binary estimate. Its value is interpretive: it asks whether the services recovery gradient strengthens as pre-event internet adoption rises, rather than only across an arbitrary high-low split (Duranton and Venables, 2015; Fujita et al., 1999).

The unemployment robustness estimate is -1.07 percentage points with a standard error of 0.25 (Bartik, 1991; Manski, 2003). The sign points toward lower unemployment in high-adoption countries after the event, but the uncertainty is wider than in the services-growth equation. That difference matters. Services output can recover through prices, hours, remote delivery, or sectoral mix before national unemployment clearly changes (Iammarino et al., 2019; Rodriguez-Pose, 2018).

5.3 Robustness

The pre-trend placebo coefficient is -0.08 with a standard error of 0.18 (Arkhangelsky et al., 2021). A near-zero placebo is not a magic wand, but it changes the conversation. It means the post-event services pattern is not immediately explained by a simple linear divergence in the years just before the shock. The remaining concern is richer: high-adoption countries may have had other time-varying advantages during recovery (Diemer et al., 2022; Barca, 2009).

The size of the services coefficient is easiest to read against the shock environment (Greenstone et al., 2010; Rosenbaum, 2002). A one- to two-point post-event services-growth advantage is not a complete recovery story, but it is material for an annual national-accounts outcome that includes many service activities with different exposure to mobility restrictions. The estimate is therefore consistent with a buffering margin: high-adoption countries may have been able to keep more transactions, work routines, and administrative processes alive while lower-adoption countries faced more complete interruption (Card and Krueger, 1994; Autor et al., 2013).

The country-year sample size also shapes interpretation (Rosenbaum, 2002; Kitchin, 2014). The regressions use 1513 observations, not millions of transactions or firms. A national WDI panel is broad and comparable, but it is coarse. That is why the paper treats the estimates as a country-level recovery pattern and does not infer firm behavior, household welfare, or within-country inequality from the coefficient alone (Monte et al., 2018; Redding and Turner, 2015).

The main economic reading is about continuity (Sun and Abraham, 2021). High internet adoption appears to mark countries where part of the services economy could keep operating or rebound faster after the common shock. The coefficient is consistent with remote work, digital payments, online delivery, and platform-mediated services playing a buffering role. It is also consistent with broader development capacity, which is why the paper does not overclaim the mechanism (Donaldson, 2018; Baum-Snow, 2007).

The continuous-adoption coefficient helps distinguish a threshold story from a gradient story (Chodorow-Reich, 2019; Imbens and Rubin, 2015). A binary split can be useful for exposition, but digital capacity is not actually binary. The positive continuous interaction suggests that the recovery pattern is not created only by comparing countries just above and below the median. It points to a broader association between the pre-event depth of internet use and the post-event services-growth path (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

The services result is stronger than the labor-market result (Imbens and Rubin, 2015; Kitchin, 2014). Table 1 makes that ordering visible. Output in services may respond before measured unemployment because firms adjust hours, productivity, prices, and task allocation before headcount. It may also reflect high-value digital services that have limited immediate employment effects. The robustness table therefore tempers, rather than repeats, the main result (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

The coefficient evidence is also asymmetric in what it can rule in and rule out (Callaway and Sant’Anna, 2021). It rules in a visible recovery difference on the services-output margin. It does not rule in a specific digital technology, a welfare improvement, or a distributional gain. It also does not rule out alternative explanations such as fiscal support, sector mix, or institutional quality. That is why the discussion treats digital adoption as a recovery capacity proxy rather than as a proven treatment (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020).

The coefficient sequence in Table 1 also shows why a different dataset changes the paper (Kline and Moretti, 2014; Gelman et al., 2020). The BDS entry paper asked whether establishment births moved in U.S. states. This WDI design asks whether national digital capacity is associated with services-sector recovery across countries. The unit, exposure, outcome, and interpretation all change. The shared structure is the discipline: define the margin, report the path, and keep causal language inside the evidence (Borusyak et al., 2022; Adao et al., 2019).

The coefficient sequence supports a bounded contribution (Gelman et al., 2020; Kitchin, 2014). Digital readiness, as proxied by pre-event internet adoption, is associated with a stronger services-growth path after 2020. The pattern is visible in the event-study figure and the static table, partly supported by the continuous-adoption check, and not cleanly translated into a precise unemployment effect. That is an interpretable finding rather than a generic digital-transformation claim (Conley, 1999; Cameron et al., 2011).

6 Discussion

6.1 Interpretation

The interpretation is that digital capacity may have worked as a shock absorber for services (Greenstone et al., 2010). When movement and contact were constrained, countries with a thicker base of internet users had more scope for transactions, work, administration, and consumption to continue through digital channels. The coefficient does not show the channel directly, but it points to a plausible recovery margin that is closer to services activity than a broad GDP comparison (Angrist and Pischke, 2009; Wooldridge, 2010).

The most important limitation is that internet adoption is bundled with development (Neumark and Simpson, 2015; Manski, 2003). High-adoption countries may have better fiscal capacity, health systems, schools, logistics, financial infrastructure, and service-sector composition. Those features could also support recovery. The analysis therefore cannot say that connectivity alone caused faster services growth. It says that the high-adoption recovery path is different after the event, and that this difference deserves a sharper design (Imbens and Rubin, 2015; Rosenbaum, 2002).

6.2 Threats

The result also has a policy warning (Chodorow-Reich, 2019). If digital capacity helps services recover, then the gains from crisis adaptation may accrue first to countries that were already connected. That can widen global differences unless low-adoption countries receive infrastructure, skills, and institutional support before the next shock. The coefficient is small enough to avoid grand claims, but it is pointed enough to make digital readiness a distributional issue (Manski, 2003; Rubin, 1974).

The unemployment check keeps the analysis honest (Moretti, 2011; Rosenbaum, 2002). A services-growth rebound is not the same as a broad labor-market rebound. Some digital services are capital- or skill-intensive, some protect output without increasing employment, and some shift work across occupations. The weaker unemployment evidence means the paper should speak about services recovery, not a generalized jobs recovery (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021).

The country-year scale leaves several mechanisms unresolved (Borgman, 2015). Tourism-dependent countries, commodity exporters, financial centers, and large domestic markets may all respond differently. A more granular design would split services industries or use firm-level adoption measures. The WDI evidence is useful because it establishes the macro pattern first, but the next empirical step should unpack the sectoral mechanisms behind it (Goodman-Bacon, 2021; Arkhangelsky et al., 2021).

6.3 Policy Implications

A second unresolved mechanism is policy response (Busso et al., 2013; Imbens and Rubin, 2015). High-adoption countries may also have had more fiscal room, faster administrative systems, or better health capacity during the shock. Those features could protect services even if internet adoption itself were only a marker. The interpretation therefore has to keep two ideas in view at once: digital readiness is a plausible channel, and digital readiness is embedded in a larger bundle of development capacity (Abadie et al., 2010; Athey and Imbens, 2016).

The event-study shape matters more than the headline coefficient (Borgman, 2015). A post-event rise after relatively modest pre-event differences is the minimum pattern one would expect if digital capacity buffered the services shock. If the event path had shown the same divergence before

2020, the interpretation would be much weaker. The figure therefore earns its place by making the timing of the claim visible (Chernozhukov et al., 2018; Wager and Athey, 2018).

The comparison also reframes digital policy (Neumark and Simpson, 2015). The point is not that every country should chase the same technology metric. The point is that broad internet adoption may be part of economic resilience because it expands the set of transactions that can survive a disruption. That is a more practical interpretation than a vague claim that digitalization is good for growth (Athey and Wager, 2019; Mullainathan and Spiess, 2017).

The policy reading is therefore about preparedness rather than ex-post rescue (Austin et al., 2018; Gelman et al., 2020). Digital adoption before the shock is what defines the exposure, so the estimate cannot be read as the payoff to emergency digitization after the fact. It points instead to an installed base of users, firms, payment rails, and administrative routines that existed before disruption. That is why the result speaks to resilience capacity: the relevant investment had to be in place before services were forced to reorganize (Gentzkow et al., 2019; Grimmer and Stewart, 2013).

The evidence is strongest as a screening result for better causal work (Borgman, 2015). A policy design could use broadband roll-out rules, mobile-network expansion, cable landings, or regulatory timing to isolate changes in connectivity more cleanly. The WDI panel cannot supply that variation, but it shows where the empirical payoff might be: services growth around a large common shock (Gelman and Hill, 2007; Gelman et al., 2020).

6.4 Conclusion

The conclusion is not that internet adoption single-handedly caused services recovery (Bartik, 1991; Manski, 2003). It is that countries with higher pre-2020 internet use show a stronger services-growth path after the shock in a transparent WDI country-year design. That pattern is economically meaningful because services were the sector in which digital substitution was most likely to matter quickly (Kitchin, 2014; Borgman, 2015).

The result changes the substantive question (Borgman, 2015). Instead of asking whether digitalization raises growth in general, the evidence asks whether pre-existing digital capacity helped preserve a specific sectoral margin when ordinary service delivery was disrupted. The answer in this build is cautiously positive for services growth and more muted for unemployment, which is exactly the distinction the interpretation needs (Anselin, 1988; LeSage and Pace, 2009).

The research implication is clear (Busso et al., 2013). A sharper design should leave the country-year WDI surface only when it can add better timing or sector detail. Broadband roll-out, platform access, firm adoption, or industry-specific services data would test the mechanism more directly. Until then, the responsible claim is a bounded one: digital readiness is associated with faster services recovery, but the causal channel remains to be pinned down (Openshaw, 1984; Fotheringham et al., 2002).

That bounded claim is still useful (Greenstone et al., 2010; Rosenbaum, 2002). It tells a reader where the evidence is strongest, where it weakens, and what kind of dataset would move the paper from comparative recovery evidence toward a causal digital-infrastructure design. The paper therefore supplies a different empirical object from the firm-entry work: a cross-country services-recovery margin disciplined by WDI data and by explicit limits on interpretation (Pearl, 2009; Keele, 2015).

The practical conclusion is that digital readiness should be studied as infrastructure for continuity, not just as a growth input in normal times (Borgman, 2015). The WDI estimates do not settle the causal question, but they identify a sector and a shock window where the association is strong enough to justify sharper evidence. That makes the next research step concrete rather than rhetorical: find better timing for digital infrastructure, keep services outcomes disaggregated, and

test whether the same recovery pattern survives a stronger comparison design ([King et al., 1995](#); [Shadish et al., 2002](#)).

The paper also clarifies what a better answer would look like ([Austin et al., 2018](#)). It would not simply add more countries or more years; it would identify when digital access changed, who was newly connected, and which service activities could actually move online. The value of the current result is that it narrows that search. It shows that the services margin is worth studying, while keeping the causal burden for the next design visible. That is a useful boundary for the next empirical pass ([Peng, 2011](#); [Wilkinson et al., 2016](#)).

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A Data Construction

The subject data appendix reports the derived World Bank WDI country-year panel. The construction keeps the indicator codes, source URL, hash, retained country-year rows, and generated estimates while discarding raw API payloads after reproducible processing (Bank, 2026; Wilkinson et al., 2016).

Table 2: Downloaded World Bank WDI country-year panel

Quantity	Value
Country-year rows retained	1895
Countries	193
Years	2014–2023
Pre-2020 internet-adoption threshold	62.60
Raw API payloads retained	No

Notes: Counts are calculated from the derived country-year panel written during data construction.

B Supplementary Source Evidence

The supplementary source evidence records the retrieval surface used for framing and reference checks. It is not treated as a substitute for subject data, but it keeps the literature sample, access rates, and search diagnostics visible in the appendix (Priem et al., 2022; OpenAlex, 2026). The reproducibility logic follows the principle that data, code, and claims should be inspectable together rather than presented as unsupported prose (Peng, 2011; Wilkinson et al., 2016).

Table 3: Open-access literature sample

Quantity	Value
Retained open-access records	60
Records with direct document link	60
Document-access share	1.000
Publication-year cells	4

Table 4: Evidence-coverage regression estimates

Model	Outcome	Coef.	SE	95% CI	N
Annual open-access record trend	records per year	-3.000	2.121	[-7.158, 1.158]	4
PDF availability trend	direct PDF indicator	0.000	0.000	[0.000, 0.000]	60
Topic relevance and attention	log cited-by count	-8.293	5.468	[-19.011, 2.425]	60

C Supporting Regressions and Robustness

The supporting estimates expand the WDI result into the full event-study table, a continuous-adoption specification, an unemployment robustness check, and a pre-period placebo. The purpose

Table 5: Robustness checks for the evidence-coverage trend

Sample	N	Trend	SE	95% CI
all records	60	-3.000	2.121	[-7.158, 1.158]
high relevance	0	0.000	0.000	[0.000, 0.000]
recent records	7	0.000	0.000	[0.000, 0.000]
PDF available	60	-3.000	2.121	[-7.158, 1.158]

is to keep the services-growth interpretation visible without turning a country-year comparison into a stronger causal claim than the data can support (Angrist and Pischke, 2009; Wooldridge, 2010).

The appendix also separates timing diagnostics from robustness outcomes: the event path asks whether the contrast emerges after the shock, while the unemployment and placebo rows test whether the services-growth interpretation travels too far. That division follows the same event-study caution used in applied policy designs (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021).

Table 6: WDI country-year event-study estimates

Model	Outcome	Event time	Coefficient	Std. error	95% CI	N
WDI country-year event study	Services value added growth	-4	-0.025	0.490	[-0.984, 0.935]	1513
WDI country-year event study	Services value added growth	-3	1.212	0.556	[0.121, 2.303]	1513
WDI country-year event study	Services value added growth	-2	0.284	0.416	[-0.532, 1.100]	1513
WDI country-year event study	Services value added growth	0	-1.105	1.057	[-3.178, 0.967]	1513
WDI country-year event study	Services value added growth	1	3.707	0.996	[1.756, 5.659]	1513
WDI country-year event study	Services value added growth	2	0.910	0.822	[-0.701, 2.521]	1513
WDI country-year event study	Services value added growth	3	1.085	1.188	[-1.243, 3.413]	1513

Notes: Event time -1 is the omitted category. Estimates include country and year fixed effects with country-clustered standard errors.

Table 7: WDI regression robustness checks

Robustness check	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Alternative outcome: unemployment rate	Unemployment rate	High internet adoption by post-2020	-1.067	0.251	[-1.558, -0.576]	1390
Pre-trend placebo	Services value added growth	High internet adoption by pre-period trend	-0.085	0.183	[-0.444, 0.275]	759

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.